USN | | | | | | | | | | [22TCSE210]

SIXTH SEMESTER B.Tech EXAMINATION, FEBRUARY - 2025
BIG DATA ANALYTICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO	Marks
		PART – A			**5 x2=10**
1.	**a.**	What are the 4 V's of Big Data, and why are they important?	L1	CO1	**02**
	b.	What is the primary purpose of YARN in Hadoop, and how does it improve resource management?	L1	CO2	**02**
	c.	What are the different data types supported in Apache Pig?	L1	CO3	**02**
	d.	What is the primary purpose of Hive in the Hadoop ecosystem?	L1	CO4	**02**
	e.	Name two key features of Hive that differentiate it from traditional relational databases	L1	CO5	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Discuss the importance of big data analytics in business and industry.	L2	CO1	**05**
		OR			
	b.	Evaluate the real-world applications of Data Science across various industries	L2	CO1	**05**
3.	**a.**	Write short notes on **i.** Resource Manager **ii.** Node Manager **iii.** Job Tracker	1	CO2	**05**
		OR			
	b.	Explain the importance of data replication in distributed systems for ensuring fault tolerance.	L2	CO2	**05**
4.	**a.**	Explain the data flow in Apache Pig from script execution to task completion.	L2	CO3	**05**
		OR			
	b.	Explain the role of the Pig Latin Interpreter in Apache Pig.	L2	CO3	**05**
5.	**a.**	How does the Hive Meta store interact with Hive queries during execution?	L1	CO4	**05**
		OR			
	b.	Explain the role of the Hive Execution Engine in processing Hive queries	L2	CO4	**05**
6.	**a.**	Explain the components of Spark architecture and their roles in distributed data processing.	L2	CO5	**05**
		OR			

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

	b.	Discuss the role of Resilient Distributed Datasets (RDDs) in Spark and how they contribute to fault tolerance and efficient data processing	L2	CO5	**05**
7.	**a.**	Examine the role of big data analytics tools in transforming raw data into actionable insights.	L2	CO2	**05**
		OR			
	b.	How can User-Defined Functions (UDFs) be used in Apache Pig?	L1	CO4	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	Analyze the opportunities and challenges associated with Big Data Analytics in modern enterprises.	L3	CO!	**10**
	b.	Compare and contrast the characteristics of traditional databases with Big Data Analytics frameworks.	L4	CO1	**10**
9.	**a.**	Describe the key components of Hadoop architecture, including the Name Node and Data Node, and their roles in managing HDFS (Hadoop Distributed File System).	L2	CO2	**10**
		OR			
	b.	Discuss the role of the Resource Manager and Node Manager in YARN and how they work together to manage cluster resources.	L2	CO2	**10**
10.	**a.**	Evaluate the data flow in Apache Pig from script execution to task completion.	L3	CO3	**10**
		OR			
	b.	Discuss the significance of complex data types in Apache Pig, such as Tuples, Bags, and Maps, with real-world examples	L2	CO3	**10**
11.	**a.**	Explain briefly about functionality of Hive Query Language (HiveQL) in managing and querying large datasets.	L2	CO4	**10**
		OR			
	b.	Examine the strengths and limitations of Hive's architecture in handling structured and semi-structured data.	L2	CO4	**10**
12.	**a.**	Explain the role of the Spark Driver and Executors in the Spark architecture.	L2	CO5	**10**
		OR			
	b.	Demonstrate how Spark SQL and Spark Streaming can be integrated into a single application for processing both static and real-time data	L2	CO5	**10**
13.	**a.**	Explain how the MapReduce model processes data in a distributed computing environment.	L2	CO1	**10**
		OR			
	b.	Analyze the key processes in Data Science and evaluate their application in a real-world scenario.	L3	CO2	**10**